



DATA SCIENCE PROJECT

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Network Intrusion Detection

Presented by:

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Problem Statement and Motivation



Problem Statement:

- Increasing cyber attacks in modern networks
- Traditional rule-based systems fail to detect new attacks
- Large-scale network data requires scalable processing

Motivation:

- Build an intelligent system to detect malicious traffic
- Use distributed computing for handling big data efficiently

Dataset Description

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Dataset Name

- NSL-KDD Dataset
- Improved version of the older KDD'99 dataset
- Designed to remove redundancy and bias

Purpose of Data

- Used for network intrusion detection
- Helps classify:
- Normal network traffic
- Malicious (attack) traffic

Number of Features

- 41 input features + 1 target variable

Feature Types:

- Numerical Features
- Categorical Features
- service (http, ftp, etc.)
- flag (connection status)

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Data Processing

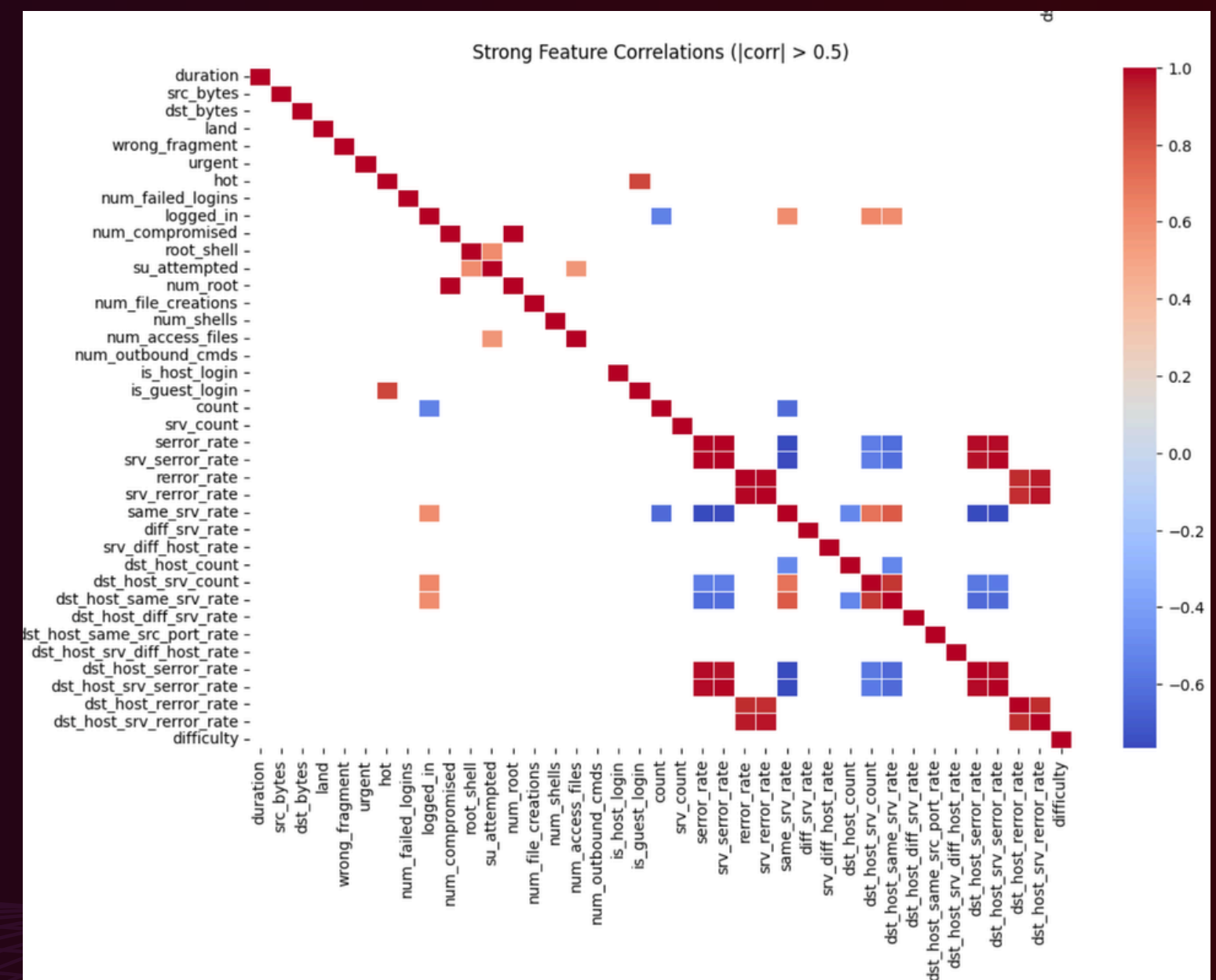
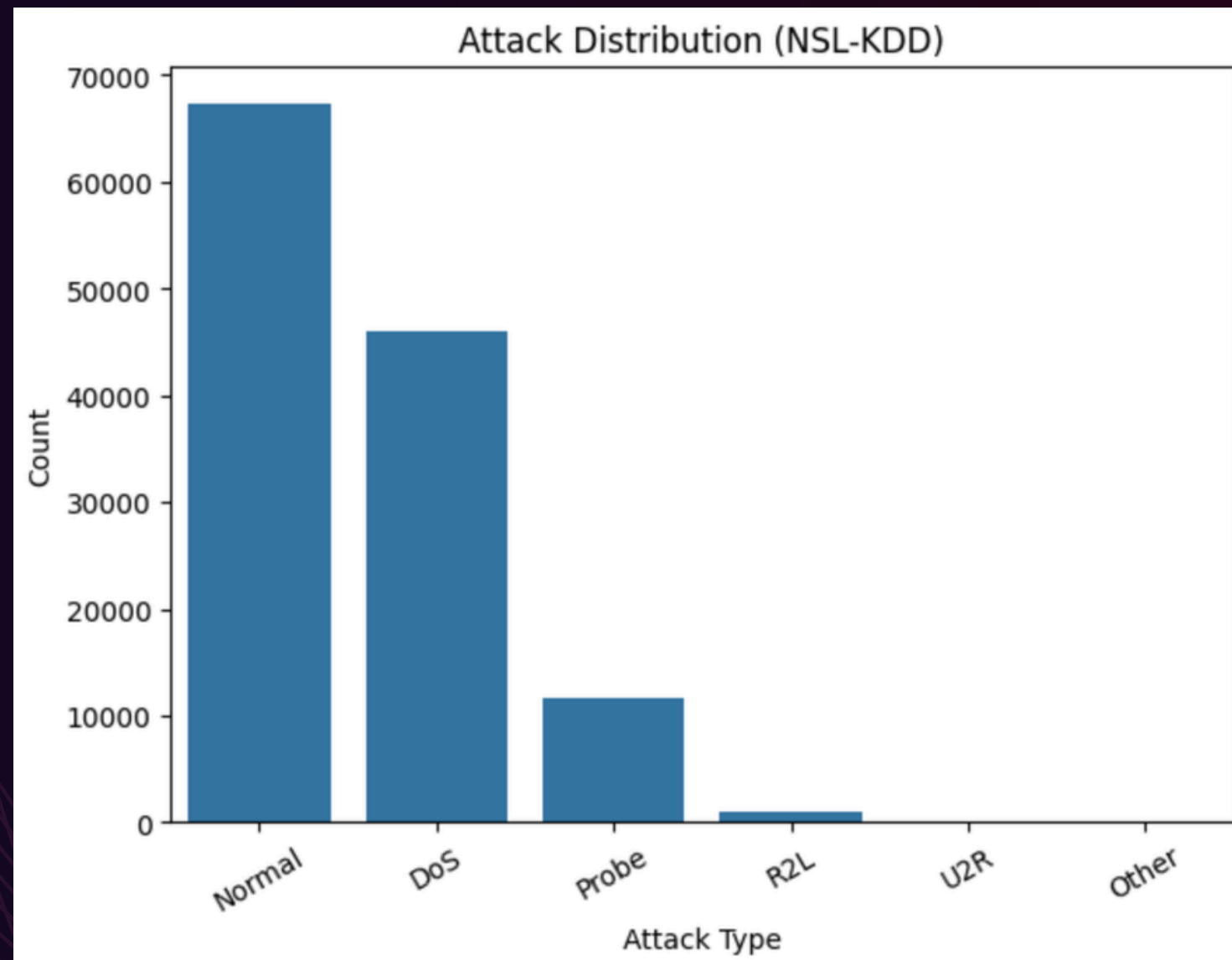
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- **Loading:** Loaded NSL-KDD Dataset
- **Inspection:** Checked structure and feature types
- **Cleaning:** Removed duplicates and checked missing values
- **Encoding:** Converted categorical features to numeric using one-hot encoding
- **Scaling:** Performed Z-score Scaling for numeric features
- **Labeling:** Converted target to binary (0 - normal and 1 - attack)
- **Distributed:** Parallel processing using Spark

Data Split:

- Training: 70%
- Validation: 15%
- Testing: 15%

Exploratory Data Analysis



Key Insights:

- Studied normal and attack traffic
- Examined relationships between dataset features
- Found key features for detection

Methodology

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Implemented using Apache Spark for distributed processing

Models used: Logistic Regression, Random Forest, Isolation Forest (Anomaly Detection)

WORKFLOW

**Data
Collection**

**Data
Processing**

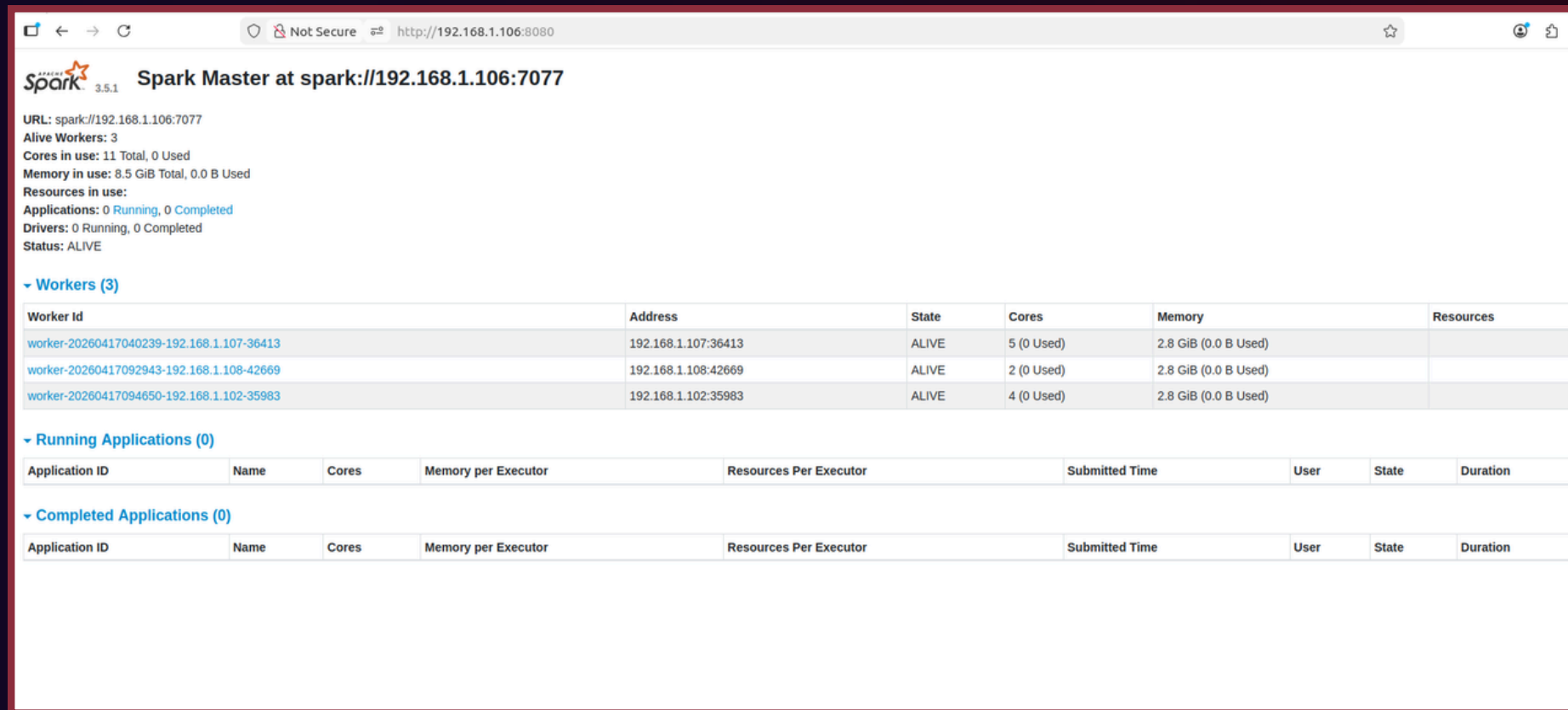
**Feature
Engineering**

**Model
Training**

Evaluation



Look How



The screenshot shows the Spark Master web interface at `http://192.168.1.106:8080`. It displays the Spark version (3.5.1) and the URL. Below this, it shows the status of the Spark Master: URL, Alive Workers (3), Cores in use (11 Total, 0 Used), Memory in use (8.5 GiB Total, 0.0 B Used), Resources in use, Applications (0 Running, 0 Completed), Drivers (0 Running, 0 Completed), and Status (ALIVE).

Workers (3)

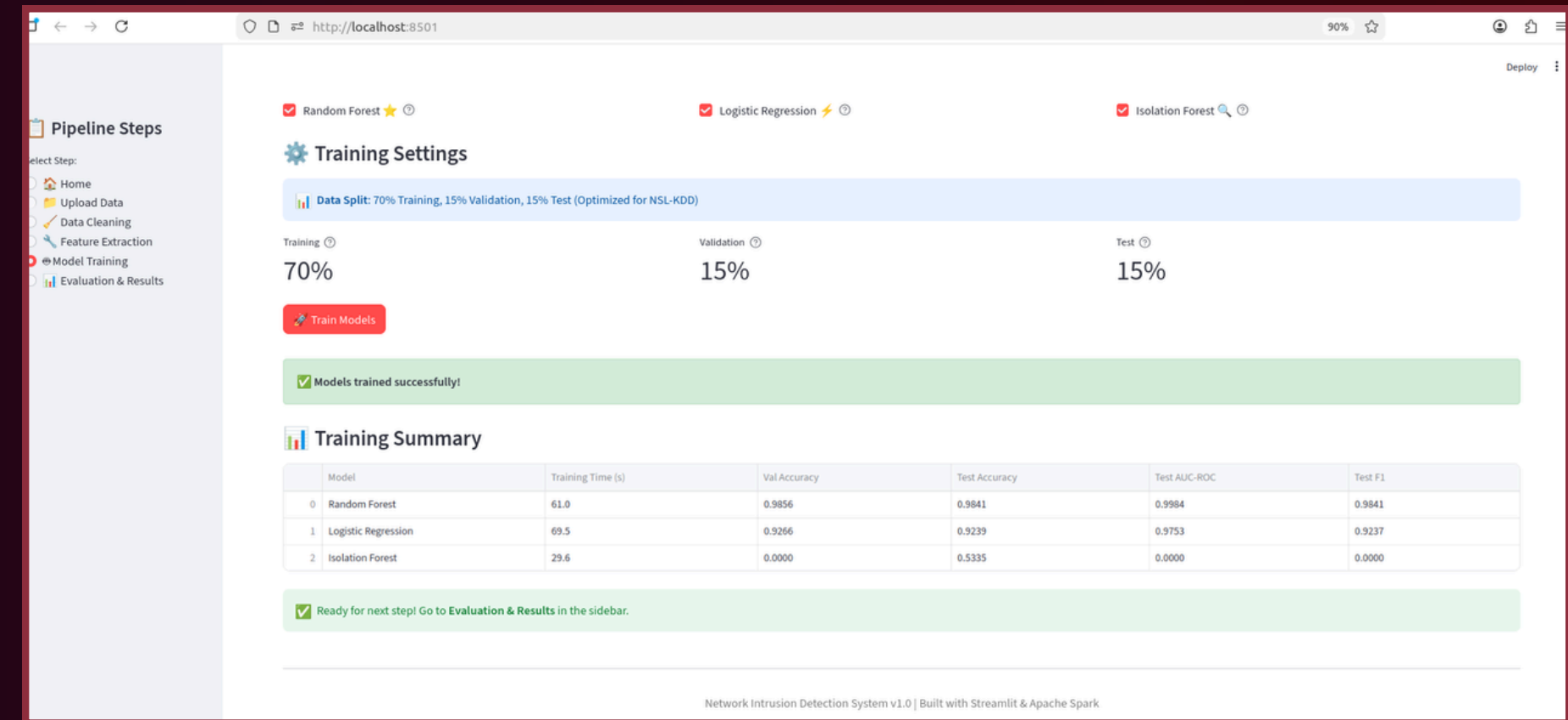
Worker Id	Address	State	Cores	Memory	Resources
worker-20260417040239-192.168.1.107-36413	192.168.1.107:36413	ALIVE	5 (0 Used)	2.8 GiB (0.0 B Used)	
worker-20260417092943-192.168.1.108-42669	192.168.1.108:42669	ALIVE	2 (0 Used)	2.8 GiB (0.0 B Used)	
worker-20260417094650-192.168.1.102-35983	192.168.1.102:35983	ALIVE	4 (0 Used)	2.8 GiB (0.0 B Used)	

Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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Completed Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
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The screenshot shows the Streamlit web interface for a Network Intrusion Detection System v1.0. It displays the training settings and results for three models: Random Forest, Logistic Regression, and Isolation Forest. The data split is 70% Training, 15% Validation, and 15% Test (Optimized for NSL-KDD).

Training Settings

Data Split: 70% Training, 15% Validation, 15% Test (Optimized for NSL-KDD)

Training: 70% Validation: 15% Test: 15%

Train Models

Models trained successfully!

Training Summary

	Model	Training Time (s)	Val Accuracy	Test Accuracy	Test AUC-ROC	Test F1
0	Random Forest	61.0	0.9856	0.9841	0.9984	0.9841
1	Logistic Regression	69.5	0.9266	0.9239	0.9753	0.9237
2	Isolation Forest	29.6	0.0000	0.5335	0.0000	0.0000

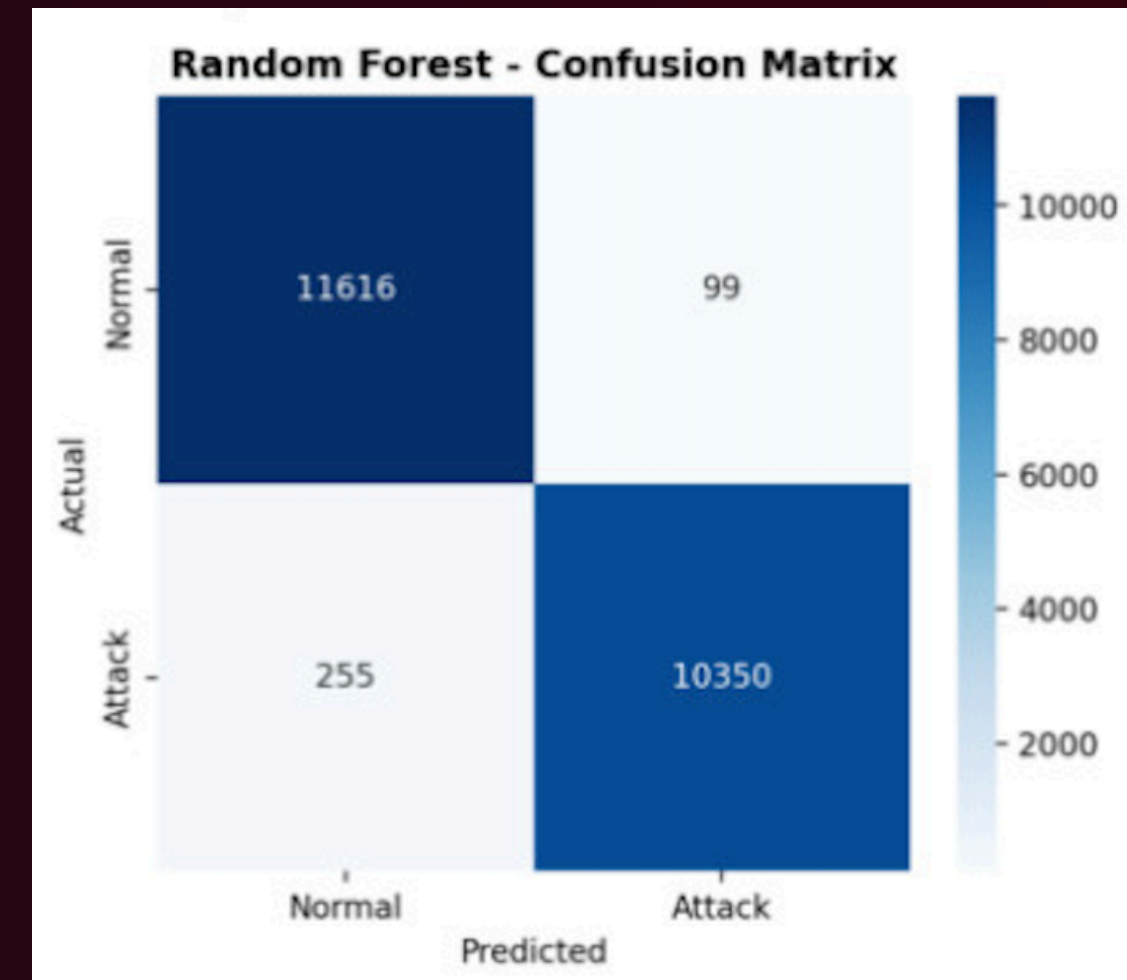
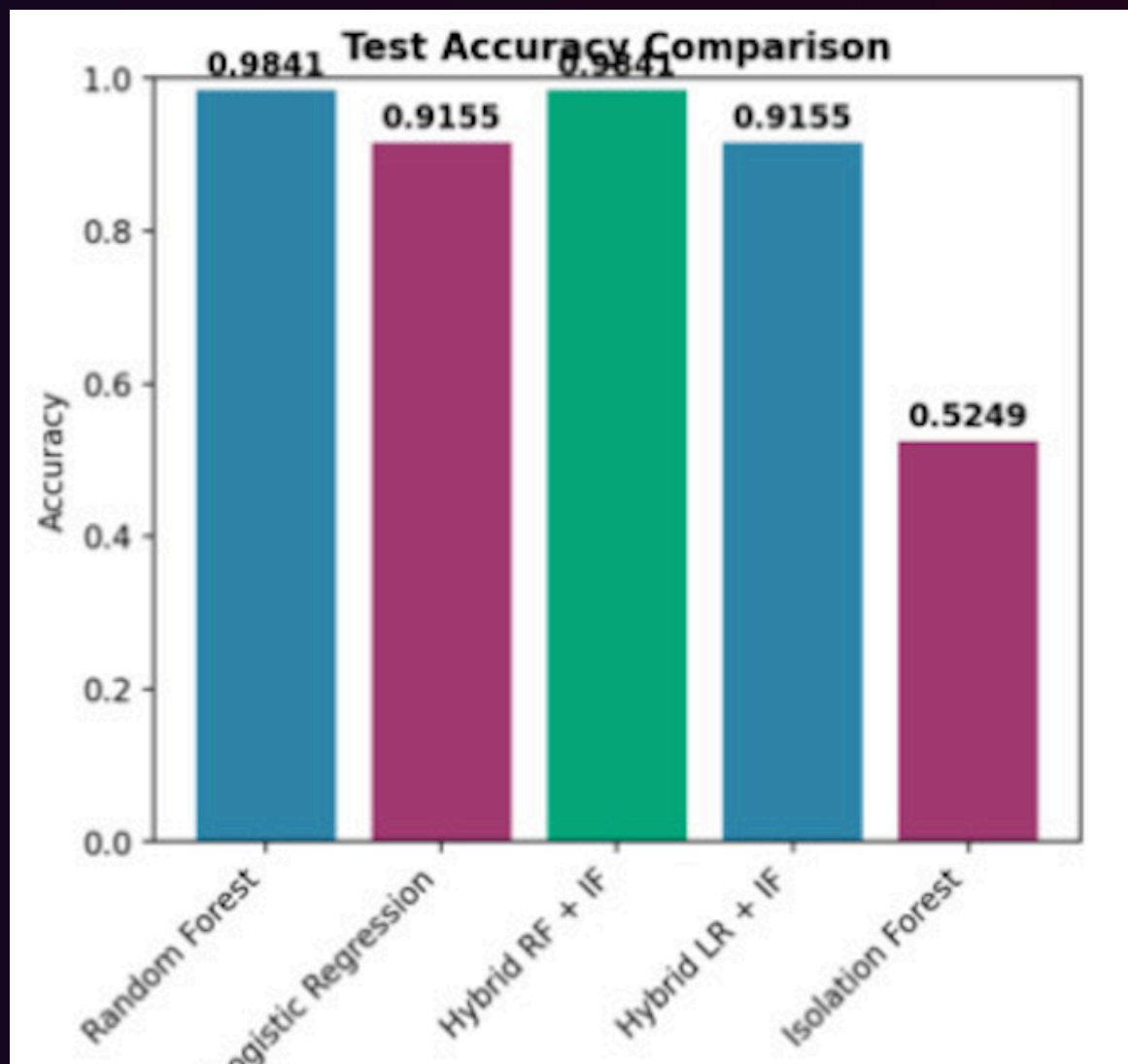
Ready for next step! Go to Evaluation & Results in the sidebar.

Network Intrusion Detection System v1.0 | Built with Streamlit & Apache Spark

This project successfully demonstrated a complete data science workflow for network intrusion detection using the NSL-KDD Dataset. By leveraging Apache Spark, we efficiently processed data in a distributed environment and built multiple machine learning models, including Random Forest and Isolation Forest. The results showed that machine learning can effectively identify malicious network traffic, while anomaly detection helps in identifying unknown attacks. Overall, the project highlights the importance of scalable data processing and intelligent models in enhancing cybersecurity systems.

Results & Evaluation

Evaluation metrics:
Accuracy
Precision
Recall



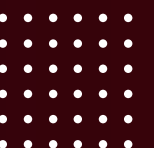
Results:

- Random Forest → ~98%
- Logistic Regression → ~91%
- Isolation Forest → detects anomalies
- Hybrid Models → improved robustness

Comparision with Reference Papers

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Approach	Technique Used	Key Findings	Limitation
Research paper - 1 ML-based IDS (IEEE 10986545)	ML-based IDS (e.g., Random Forest / Ensemble)	High accuracy in detecting known attacks	Limited detection of unknown attacks
Research paper - 2 Hybrid IDS (IEEE 11012645)	Hybrid / Advanced ML models	Improved detection performance	High computational complexity
Our Proposed Approach	Random Forest + Isolation Forest + Spark	~98% accuracy + detects unknown attacks	Not real-time



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Challenges and Limitations

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The project faced challenges such as handling class imbalance that can bias model performance, encoding high-cardinality categorical features like service, detecting unknown or zero-day attacks, ensuring efficient distributed processing with limited nodes, avoiding overfitting, and interpreting complex model decisions.



Dataset Limitations

The NSL-KDD Dataset is outdated and may not reflect modern cyber-attacks.



Not optimized for real-time detection

The model is trained on static data and may not perform efficiently in real-time environments.



Limited Generalization

Models trained on this dataset may not perform well on real-world unseen network traffic.



Resource Constraints

Distributed setup with limited machines (2 nodes) restricts full scalability and performance benefits.

Thank
You